

Archaeology of the Silk Road: Challenges of Scale and Storytelling

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ABSTRACT:

Invented in the nineteenth century as an allegory for large scale human interaction across Eurasia, the idea of 'the Silk Road' continues to shape archaeological investigations of trade, travel, cultural exchange, and mobility in the region between the Near East and East Asia. Though long used to refer to trade between the ancient and late medieval periods, the framework of the Silk Road has grown increasingly popular and is used to orient research on mobilities of much earlier periods, as well as to frame movement and exchange at the molecular level, including of human genes. This article reviews the shared challenges confronted by Silk Road archaeologists and explores the narratives about human culture that have been tied up in the Silk Road metaphor from the beginning. Through a review of recent work on and along the Silk Road, the article traces common narratives and shared scalar challenges across archaeologies of landscape, material culture, gender, mobile lifeways, and isotopic and genetic assemblages, and examines tensions between globality and locality within Silk Road cultural heritage and the implications of China's Belt and Road Initiative.

KEYWORDS:

Silk Road, Central Asia, Eurasia, gender, mobility, trade, aDNA, heritage, STS, post-colonial archaeology

STATEMENTS AND DECLARATIONS

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Introduction

This review of archaeology of the Silk Road is concerned with both the archaeological construction of narratives of interaction and exchange across ancient and medieval Eurasia, and the legacy of these narratives; or in Tamara Chin's phrasing, with "two Silk Roads: not the ancient and the modern, but the invented and the reinvented" (2013, p. 194). The evocative and popular term 'the Silk Road' refers to the historical development of cultural connectivity across Eurasia, and the ramifications of this connectivity on world history. Scholars writing about the Silk Road emphasize the co-construction of Eurasian geographical and ecological features— steppe pastures, river systems, and 'corridors' through mountains and deserts— with networks of settlement, urbanism, and movement: of people, goods (including silken textiles), cultigens, technologies, religions, genes, and languages. Conventionally, the earliest period of Silk Road trade is set in the first centuries BC, with a focus on the contacts and trade links enabled by expansion of Hellenistic, Roman, Parthian and Han Chinese empires. Other high points of Silk Road cohesion include the eras of Sogdian trade connecting the Islamic caliphates, Byzantine Empire and the Tang and Song empires of China (sixth-10th centuries AD), as well as the brief 'pax Mongolica' following the thirteenth-century Mongol conquests. Conservative arguments as to the spatial extent of Silk Road trade might focus on a single region, while more expansive models stretch from China's port cities to the Viking colonies of North America (but see Figure 1 for a focus on Eurasia). Likewise, temporal spans range from the millennia invoked by claims of 'paleolithic Silk Roads' to the few decades' duration of the medieval Mongol *paiza* passport system (Blair 2005; Iovita et al. 2020).

The Silk Road has been variously conceptualized: as a shared culture made possible by the spread of nomadic political mythologies (Beckwith 2009), as emergent tangled threads of shared visual and material culture carried by diverse practices of mobility (Whitfield 2015), or as a mobile, material politics centered on textiles, patronage and moral-visual language (Allsen 1997). Some archaeologists are concrete and literal in their definitions of the Silk Road as a system of routes, while others define it more generously as a phenomenon of pre-modern globalization. Nearly every discussion of the Silk Road includes two observations: that the thirteenth-century Venetian traveler Marco Polo was the exception to the rule that few people ever traveled a 'road' from Europe to China; and that the term 'the Silk Road' was itself invented in the late nineteenth century (Chin 2013; Khazanov 2021; Waugh 2007; Whitfield 2015). Conversation among archaeologists who deploy the 'Silk Road' in titles and abstracts frequently touches on the utility of the term for situating the history and archaeology of diverse parts of Eurasia within a recognizable paradigm. The boom of interest in the Silk Road has been accompanied by a parallel boom in writing pointing out the devalued status of the term as, at best, an uncritical "bandwagon and gravy train," and at worst, an orientalist holdover that preempts actual knowledge-making about Eurasian pasts (Ball 2019; Rezakhani 2010). As this article reviews, the idea of the Silk Road remains contentious even as it generates new research questions or is proposed as a methodological orientation in itself (Knutson 2021). The concept of the Silk Road has enabled archaeologists and historians to frame case studies in the premodern and non-European past as comparisons with and challenges to modernist understandings of globality, globalization and deterritorialization; at the heart of these

considerations are complex, recurring, and interwoven challenges of *scale* and *storytelling* at every stage of research from design to dissemination.

In this brief review, I consider where the framing narratives that orient work on the Silk Road have come from and review the broader context for ongoing archaeological research in Eurasia. My own orientation is more methodological than strictly geographical: rather than present a review of new archaeological material pertinent to the world of the ancient and medieval Silk Road, which is equivalent to the entirety of Eurasia, I am going to center work that deploys or invokes the Silk Road concept explicitly, and examine what work that concept does, what it enables, what it conceals. I am curious about the question of the agency that we give to the Silk Road in explaining change in the past, especially as juxtaposed with the question of what (if any) role global imaginaries played in the everyday lives of people living between Barcelona and Beijing. This discussion is interdisciplinary by necessity; as I explore, bridging the scalar gaps in datasets, analyses, and phenomena pertaining to the Silk Road requires a parallel spanning of interpretive approaches and methodologies aimed at material ranging from the monumental to the molecular. A thread running through the discussion is also the fundamental and inalienable politics of archaeological knowledge-making about the Silk Road: like all archaeology, work in this region is intimately tied to colonial and imperial projects of the past and present, and to tangled politics of agency, self-determination, and identity. This entanglement is inherent to the framing of the Silk Road as a narrative and scalar imaginary, with the result that deployment of the concept within archaeological work brings ethical obligations on the part of the discipline. My approach is situated across a few intersecting

theoretical concerns shared by anthropological archaeologists working on Silk Road questions, which for clarity I summarize here.

The first orientation is around *post-globalization critiques of **scale***: Anthropologists critical of globalization have repeatedly highlighted the challenge of working across vast disparities of actual and imagined scale (Massey 2013; Tsing 2004; Werbner 2008). The challenge of framing causality, contingency, and agency is exacerbated by considerations for perception and imagination on the part of practitioners, and for the fact that effective (acting globally) and affective (feeling global) participation in global cultures are not the same and are not evenly distributed. These interventions critique the reductive model of participation in large scale phenomena (like the Silk Road) that emerged from twentieth-century discourse on globalization, which frames such participation as the impact on and/or transformation of the local, traditional, and frequently subaltern by a universalizing, 'modern' hegemonic global. For archaeologists of the Silk Road, this critique means a freighting of our attention to the modes in which people in the past constructed, participated in, perceived and represented cultures and worlds at different scales.

My second orientation is through *post-colonial and feminist critiques of **storytelling in science***. This review is partially oriented through science and technology studies (STS) as a consideration of the cultures of knowledge production in archaeology, with a special attention to the use of *metaphor* and *narrative* to bridge gaps in our data and in our understanding. My approach draws upon arguments within social and material history that posit that we share with the people we study a practice of understanding our past and present in narrative ways (e.g. White 1973; Williams 1975; Zemon Davis 1987). The Silk Road is a story about human

desire and encounter that itself produced spatial models mapped onto the landscapes of the 'Silk Road lands,' shaping in turn the contours of our datasets and the forms of the stories we craft with those data (for a longer discussion see Franklin 2021). My attention to the Silk Road as a narrative device also draws on the discourse influenced by Said's *Orientalism* (1978), which is attuned in particular to the role of humanities and sciences, including archaeology, to the constructing of geopolitical-cultural narratives centered on the encounter and mutual (if not equal) desire of an 'occident' and an 'orient.'

Knutson recently picked up on the zeitgeist of Silk Road archaeology, summarized in her observation that "the Silk Road is not only useful as a metaphor but also as an *analytical model*" (2021, p. 2; emphasis in original). What happens when we allow a slippage between metaphors and models in archaeology? Both are mechanisms of making knowledge about the world: metaphor helps us understand complex and abstract concepts in material—but always cultural—ways, while models are deployed as confirmable narratives of causality. Both are critical for science; their tendency to cleave to one another underscores the necessity for ongoing reflections on the cultures within which we do archaeology. A specific methodological question I ask repeatedly is, to what extent does the shorthand-story of the Silk Road serve as a 'black-box' for archaeological models of cultural change in the past? The term 'black box' comes from cybernetics, and refers to a placeholder drawn in systems diagrams whenever the internal workings of a particular machine are too complex to incorporate into the process under discussion. As Bruno Latour wrote of such complex apparatuses: "no matter how controversial their history, how complex their inner workings, how large the commercial or academic networks that hold them in place, *only their input and output count*" (Latour 1987, p. 3; my

emphasis). As I explore, the idea of the Silk Road seems to be a useful shorthand for cultural, ecological and material complexity over vast spatiotemporal distance; however, the term can also black-box the articulations of contingency and agency that allow for differences in human experience and critical nuances of archaeological explanation, reducing these to an 'input' (encounter) and an 'output' (cultural change) along a simplified east-west axis.

Of equal importance among the questions that can be obscured by this black-boxing is that of the culture and politics that shape both the creation of scientific data and their interpretation. The attractiveness of the Silk Road as a metaphor has emerged as a critical issue within twenty-first-century geopolitics and the bundling, in turn, of archaeological and cultural heritage as a technology of nation-making. I examine the impact on archaeological knowledge communities of the congelation of the Silk Road as a category for UNESCO World Heritage designation, and of the announcement in 2013 of China's Belt and Road Initiative. The re-elevation of the Silk Road as an explicit interest of geopolitics has resulted in new opportunities for research, but also an increased urgency in the need for archaeologists to reflect on the links between our work and the projects of modern nation-states and corporations. The conceptual constellation of Silk Road heritage, archaeological research and the interests of the nation-state also adds urgency to a consideration of the participation of Silk Road archaeology within the broader, ongoing archaeology of gender and embodiment. As has been effectively argued by critical anthropology of globalization, gender is always present in perceptions of globality and locality, and implicated in the accommodation of continuity and change in everyday practice (e.g. chapters in Werbner 2008). My ultimate aim in this article is reflection rather than

polemic; more critical thinking about the work that ‘the Silk Road’ model does will germinate richer as well as more responsible science.

Historical Background

To understand the relationships between themes emergent in this interdisciplinary research and their implications for public understanding, heritage politics and transregional nation-making, it is necessary to back up and examine the roots of the archaeological construction of the Silk Road. By now it is a narrative commonplace for work on the Silk Road to begin with a nod to the nineteenth-century construction of the term, which is universally attributed to Baron Freiherr von Richthofen (to cite a few: Chin 2013; Frankopan 2015; Knutson 2021; Latov 2010; Mair and Hickman. 2014; Spengler 2020; Whitfield 2007; Wood 2002). As with many concepts that orient science, these references to the nineteenth-century coining of the Silk Road term speak to the challenges of working with entities that must be *constructed* even as they are *discovered*. But the idea of the Silk Road transcended historical utility and gained archaeological reality across the turn of the twentieth century with a wave of expeditions, excavations, and intensive collecting projects undertaken by Europeans in Central Asia, South Asia, and China (Hansen 2012; Whitfield 2020; Wood 2002). Of the numerous expeditions, the most notorious were the travels in the Tarim Basin and Xinjiang by the Hungarian-English polymath Sir Marc Aurel Stein (Stein 1903; Stein 1946) and Swedish explorer-geographer Sven Hedin (Hedin 1938). During a period when western publics were hungry for the discovery of ‘lost’ civilizations, and scientists eager to further orientalist quests for the origins of western modernity in the linguistic and civilizational antiquity of Asia, Aurel Stein’s expeditions in particular provided popular accounts of cities buried in the desert sands of the Tarim Basin (See

Figure 2) and furnished an enormous array of objects for museums. Of course, the artifacts and architectural fragments— including textiles and wood in astounding states of preservation— were collected as a by-product of the search for texts, and of the thousands of written documents that Stein recovered, the most famous are the archives of the caves of Dunhuang. Stein’s textual assemblages fueled the general mania for discovering lost and unknown languages in their ‘homelands’ of Central and South Asia, and the desire in particular to locate the wellspring of Indo-European (IE) languages (and the invented ‘Proto-Indo-European’), and in particular ‘Tocharian,’ the name attached to a branch of IE recorded exclusively on texts from the Tarim Basin (Mallory 2015, p. 3). It was in part from this broad discourse that an imaginary emerged of the Silk Road as a transporter to the West of civilizational transformation *ex oriente*.

The published works of Stein and his contemporaries express the complex status of archaeological research during this period in the regions of east Central Asia. Stein and Hedin were working with the consent of the Chinese state, which was at that time in the process of negotiating the incorporation of western regions under its control. The success of their expeditions, and in particular the permission to excavate and collect material (that they did not always obtain) were contingent on negotiating the patchy bureaucratic presence in the western regions, and the progressive suppression of local Uyghur resistance against Chinese hegemony. Numerous rebellions were in turn linked to first Russian and then Soviet interventions in Central Asia, a dynamic referred to in the nineteenth century and up to the present as the ‘Great Game.’ As Hedin recounted in the early pages of *The Silk Road*, his 1933 expeditions were given free rights of movement and protection in order to “revive the old Silk Road and

turn it into a modern motor road,” in service of the aim of quelling Uyghur resistance and edging out Soviet influence (Hedin 1938, p. 13).

During the late nineteenth and early twentieth centuries Russian and Soviet archaeologists also explored the regions of the broader ‘Silk Road world.’ While the pre-war work of scholars like Piotrovskii and Marr would draw the Caucasus into broader Eurasian models of migration and cultural contact (Marr 1934; Piotrovskii 1949), Soviet expeditions to Central Asia generated new data on ancient settlement, the transformation of landscapes, and the articulation of transcontinental trade networks. Writing in this period frequently frames Eurasian exchange through a Marxist focus on exterior trade as a stimulus to growth rather than as a phenomenon itself; Tolstov for instance does not use the term ‘Silk Road’ in the synthetic publications of the Khorezmian Archaeological-Ethnographic Expedition (Tolstov 1948 a and b; see discussion in Stride et al. 2009). Soviet expeditions through the twentieth century continued to follow imperial footsteps; perhaps most famously, the work of Victor Sarianidi and the South Turkmenistan Archaeological Complex Expedition (YuTAKE), resulting in the discovery of the cosmopolitan burials at Tillya Tepe, represented the co-incidence of archaeological research and (Soviet, British, French, American) imperial border-making of Afghanistan (Sarianidi 2005, see also Hiebert 1994). Soviet archaeology shared a basis in orientalism with the social sciences of rival imperial powers; also, Soviet research in particular was informed by the cultural-political arguments of ‘Eurasianism,’ an intellectual movement that argued for the steppic roots of Russian culture (Fabian 2020; Halperin 1982).

Reorientations: Silk Road Archaeology in the Post-Empire Era

Ongoing archaeological research under the rubric of the Silk Roads has developed on the foundations of collaborative international work started just before and immediately after the collapse of the Soviet Union, as archaeologists from the former 'western bloc' have developed collaborative relationships with scholars working in formerly Soviet academies and newly coalescent institutes from Hungary to Mongolia. On one level, the Silk Road as an orienting concept has proven to be one solution for the under-representation of Eurasian archaeology within disciplinary conversations and public interest (c.f. Hanks 2010). For the first decades of what is still often called Eurasian archaeology, it was frequently observed that a continent-scaled disciplinary attention could be attributed both to the availability of Russian as a lingua franca, as well as a shared interest among anthropological archaeologists, art historians, and linguists with the lifeways and cultures of mobile pastoralists and, in turn, with the 'sea' of grassy steppe that links the Carpathians with the Tian Shan.

Meanwhile, curated assemblages that demonstrated the interchange of cultural influences across Eurasia from the Bronze and Iron Ages up to the early modern period had been circulating in the US and Europe since the latter decades of the USSR. Perhaps unsurprisingly, the narratives around these collections mirrored the binarisms foremost in Cold War rhetorics, even as the exhibitions themselves aimed at dismantling civilizational enmities. In his introduction to the 1978 catalog of the blockbuster exhibit of Scythian metalwork from the Hermitage at the Metropolitan Museum, director Thomas Hoving mused that when looking at the style and form of Scythian figural work, "one feels as if one were seeing the archetypes of the two fundamental styles of art: one, the abstract decorative art of hieratic symbolism that

will wend its way through time, through Byzantine art, through the mystifyingly complex animal interlaces of the Book of Kells, and even to the abstract influences of our day. The other reflects the serene, balanced human idealism of the ancient Greek experience” (Hoving 1975, p. 1). This entwining in the steppe heartland of two diverging and yet complementary civilizational impulses—western/humanist and oriental/primitive—set the dualist tone for the increasing numbers of exhibitions on artifacts of the Silk Road worlds. In the last decade or so however, Silk Road exhibitions have proved a deft method for using museum collections to think through the implications of globalization, to play with the categories that structured traditional curation, and to kindle fascination with the complexity and rhizomatic un-boundedness of past cultures. Exhibition spaces have also consistently furnished a literal and discursive space for expansions upon the Silk Road model; for instance, the symposium associated with the 2011 Penn Museum exhibit *Secrets of the Silk Road* generated an early call to extend the reach of the Silk Road-as-metaphor to cover prehistoric interaction, mobility and exchange in Eurasia (Mair and Hickman 2014). Among the artifacts in the *Secrets of the Silk Road* exhibition were mummified human remains from the Tarim basin, on loan to the traveling show from the Xinjiang Autonomous Region and the Xinjiang Institute of Archaeology in Urumqi; the exhibit of these remains was suddenly curtailed in 2011 by orders from Beijing, related to the increase in tensions surrounding China’s ethnohistorical claims to Xinjiang, a region argued at the time to be an “inalienable part of China” in the exhibits of the Urumqi museum (Rothstein 2011; see also Wong 2008).

Public awareness of the Silk Road has skyrocketed in the last ten years due in particular to two linked events: the 2013 announcement of China’s Belt and Road Initiative (BRI, also

called the One Belt One Road or New Silk Road Initiative), and the 2014 designation by UNESCO of the Chang'an-Tian Shan Corridor of the Silk Road as a World Heritage Site. In a 2013 speech at Nazarbayev University in Nur-Sultan (then Astana), Chinese President Xi Jinping named himself as a native of Shaanxi province, the "starting point of the ancient Silk Road" (Xi 2013). Remarking that he could "hear the camel bells echoing in the mountains and see the wisp of smoke rising from the desert," Xi rhetorically compared himself and his Initiative to the "mission of peace and friendship" of the Han dynasty envoy Zhang Qian, whom he credited with opening the Silk Road in the second century BC (Xi 2013). The UNESCO designation itself represented the result of more than two decades of research and discussion that emphasized the Silk Road as an engine of diversity and demystification, enabled by "east-west contacts" and "two-way transmission" of trade, technology, and culture. The linked phenomena of UNESCO World Heritage and BRI, which is discussed further below, are the context for the continuing expansion of interest and support for transregional research in Eurasia and beyond (c.f. Storozum and Li 2020; Whitfield 2020).

Stories of the Silk Road

One of my aims in the above review is to establish the historicity of two primary narratives related to the Silk Road. These 'Silk Road stories' are frequently, and often inadvertently, invoked when the Silk Road is used as a convenient black box for the complexity of human, material, and ecological interactions across space and time.

The first of these stories is that *of the Silk Road as a continent-spanning mutual regard of 'east' and 'west.'* In this narrative framework, the Silk Road region, and especially the territories of what is now western China and Central Asia, are approached as a 'meeting point'

of cultures, and specifically for bounded concepts of ‘western’ or European culture, and ‘eastern’ or Asian culture. These cultures are frequently further bounded within opposing states, such as Rome and Han China, or Byzantium and Tang China. These mutual fascinations are drawn from (sometimes wishful) readings of historical texts; however, as historians have argued, rather than an empirical fact about the world, the idea of a mutual fascination between west and east was a geopolitical imaginary that was strenuously constructed on the part of ancient polities (Canepa 2009; Rezakhani 2010). The embeddedness of archaeological research such as Stein’s within imperial ambitions reveals the space of the Silk Road as a locus for the dialectical construction of nineteenth- and twentieth- century empires in opposition to Asian others, even as they traced the origins of European hegemony from orientalist roots.

The second of our persistent Silk Road stories is the narrative of the Silk Road *as a hero’s journey*. As we see in Xi Jinping’s speech from 2013, there is a tendency to embody the encounter of opposing cultures within heroic historical travelers, such as Zhang Qian, Zheng He, Alexander the Great or Marco Polo. These narratives in turn evoke the Silk Road as a space of adventure, of encounters experienced by (usually male) mobile, agentive individuals. As Peter Brown stated it: “In this pre-modern pattern of globalization, great men and their women set themselves off against the all-pervasive drabness of ordinary life—whether in the village or on the steppes—by prizing difference: difference in goods, learned servants, women, and even animals” (2014, pp. 17-18). Note the distinction made here between the phenomenon of the Silk Road, which is separate from and impacts the realm of everyday life (e.g. the context of much archaeological data), as well as the opposition between active masculine mobility and

passive sedentism. These heroic figures thus serve to embody the mobile global that impacts, transforms, demystifies, conquers, seduces, or assimilates the traditional local.

Both of these narrative shorthands pose challenges to archaeologists, especially when they creep into our analysis— or in other words, when the Silk Road slips from metaphor into method. I return to these through the discussion that follows as recurring tropes that can be shortcuts or obstacles in archeological reconstructions of scalar participation and social agency in the past. The appeal of these tropes is understandable given the methodological challenges faced by archaeologists to insert human-scale agency into the roaming contingency of continent-scale phenomena; it is with these challenges in mind that I dive deeper into the diverse research making up the ongoing archaeology of the Silk Road.

Deterritorialization and Premodern Globality: Silk Road Archaeology in Practice

Archaeological and historical research in recent decades has drawn on terminology and conceptual tool sets from parallel negotiations of scale, contingency, and distributed agency among anthropologists and historical archaeologists working on modern globalization. Emergent from this is a useful label for archaeological strategies for tracking space-time systematics in material culture in the context of large-scale mobility: we study phenomena that are *deterritorialized*, cultural ideas and technologies that take to the road, which can't easily be tied to singular landscapes or unique local 'traditions' (Deleuze and Guattari 1987). And yet archaeologists continue to work *in place*, engaging with excavated and curated assemblages and architectures, as well as working *on places*, tracing the *longue durée* co-construction of people and their landscapes. This raises a question of whether 'the Silk Road' remains effective

as a scale of engagement; does anyone work, live, make politics or otherwise dwell at the scale of the global? Or are we constantly also encountering, tracing, and inventing practices of *re-territorialization*, embodying the global into the matter of tradition, memory, locality, authenticity, heredity? I examine methodological navigations of these challenges through a number of lenses at overlapping scales; from artifacts, to pastoral lifeways, to molecular assemblages, to constellations of material culture, memory, and ongoing practice within Silk Road heritage. I call attention to the recurring invocation by archaeologists and their collaborators of the narrative metaphors described above, as they work to craft a method capable of bridging disparate scales of time, space, and agency.

The Silk Road Lives of Things

In an episode during the excavations at the Tarim Basin site of Niya, Sir Marc Aurel Stein described coming upon a collection of Hellenistic seals, found on documents preserved by the arid sands of the Taklamakan desert. Stein marveled at the freshness of the impressions; he remarked on the parallel between the sealing of ancient documents with these impressed classical images, and his own act of sealing the covered excavation itself “with [his] electrotpe reproduction of a beautiful gold coin of Diodotus, the first Bactrian Greek King” (Stein 1946, pp. 283-4). In recounting the romantic shock of recognizing transported objects so similar to his own personal seal (which would later appear on all of his published works), Stein collapsed the spacious narrative of Silk Road travel and trade into an intimate encounter with particular artifacts, casting himself and the seals as fellow travelers. A major thread running through archaeology of the Silk Road, from Stein onward, is the effective narrativization of seemingly out-of-place objects, which contain in the spacetime systematics of their production and

recovery contexts a silk-road-in-miniature. Much of this work has been done by curators and collections managers; for instance, Susan Whitfield and others have established a mode of storytelling anchored in specific objects that are, through typological and art historical analysis, linked to traditions situated along the silk routes or at either 'end' (Bickford Berzock et al. 2007; Whitfield 2015; Whitfield 2018; Wu, J., 2004). Research in the same mode has expanded the list of things transported on the Silk Road to include more ephemeral technologies such as musical instruments and performance (Millward 2008); the array of practices linked to growing, cooking and eating food (Baker Brite 2021; Jaffe and Flad 2018; Spengler 2020); or cosmologies of the body linked to objects like mirrors (see discussion below; Yao 2012; Zhangsun et al. 2017; and Frachetti and Bullion 2018 for prehistoric comparison). As Whitfield remarked in the introduction to her 2018 *Silk, Slaves, and Stupas*, traveling objects are frequently made into heroic 'great men' (2018, p. 2); however, Whitfield and others provide counternarrative through close reading and juxtaposition, effectively crafting Silk Road microhistories motivated by encounters with de-stabilizing, boundary-crossing things.

A parallel technique is the archaeological and art-historical focus on the movement not of objects themselves, but of styles and technologies of making, deterritorializing the cultural from the material. An example of this discussion is concerned with the motifs and images reproduced on textiles across the Silk Road world and from the classical period onward. Symmetrical motifs of animals, trees, hunters, and mythical creatures 'migrate' across objects, including portable things like textile fragments, ceramics, and figurines, as well as immobile statuary, art, and architecture (Hoffman 2001; Takeuchi 2009; Whitfield 2018; Żuchowska 2015). A major category pertains to motifs and styles linked to the religious communities that

expanded through Eurasia in the late antique and medieval periods, whether the westward spread of Buddhism (Abdullaev 2000) or the eastern movement of Zoroastrianism (Cheng 2010). An enormous discursive space is also occupied by the category of Gandharan art, or statuary and numismatic depictions from what is now northern Pakistan and eastern Afghanistan that seem to combine classical and 'eastern' traditions of representing human bodies in particular (Khettry 2011; Morris 1983; So 2006). The nineteenth-century fascination for Gandharan or 'Greco-Buddhist' art was driven by paired desires on the part of British imperial scholars, "to be Alexander and to find Alexander" (Mairs 2018, p. 577); the resulting art historical entity has in turn been used to "politically justify and culturally stabilize" the identities of both colonial and post-colonial nations (Falser 2015, p. 3). The question of Gandharan art and its possible relation to an ancient political/ethnic group such as the Kushans or Yuezhi also touches on a number of methodological challenges fundamental to Silk Road archaeology: a necessary reliance on objects in museum and private collections with unknown provenance due to imperial collecting practices (Stewart 2016), ongoing looting of findspots due to continuing unrest in source-countries such as Afghanistan (Stein 2015; Van Krieken-Pieters 2006), and slender correlations with textual sources (Canepa 2010). Ultimately, a core challenge is the tendency for re-territorialization of mobile, partible Silk Road influences and trans-border cultures, whether into stable ethnicities of the past or origin narratives of nation-states of the present.

Silk Road Places: Local Projects with Global Contexts

As with broader archaeologies of global processes including globalization, engagement with the Silk Road as a paradigm has provoked increased attention to not just the impact of the large

scale on the everyday, but also the perception of and creation of conditions of possibility for world-scale cultures in locally situated practices. At a basic level, this manifests in the interpretive discussions of excavation results, moving the locus of significance from a narrow view on a local 'cultural group' to a consideration of how more remote settlements and frontier societies participate in cultural and technological transformations previously studied primarily in urban contexts.

Work on ancient and medieval Eurasia also increasingly challenges the very categories of 'urban' and 'frontier' as orienting principles for the study of social life, architecture, infrastructure and cosmopolitanism. These reorientations have manifested in increased attunement to the broader networks of mobility, exchange, and settlement that contextualize the urban sites listed on the itineraries of travelers like Marco Polo and William of Rubruck, neither of whom had a particularly close eye for extra-urban settlement. For instance, in 2010 Brisbane et al. published a collection of studies of the broader settlement network of Veliky Novgorod, the center of the Kievan Rus. This interdisciplinary research on architecture, material culture, and environment demonstrates that trade links in the medieval Volga drainage were sustained by landscapes of settlement, not single roads. Assemblages from the rural site of Minino in particular show Baltic trade links, mirroring the finds at medieval Scandinavian settlements like Birka of Persian silks transported up the Volga routes (Vedeler 2014). Also in 2010, the Mongolian-German Karakorum Expedition (M-GKE) published a first volume of interdisciplinary research on the medieval Mongol capital city and its excavated contexts and assemblages (Bemmann et al. 2010). The core of the study is a consideration of the 'craftsmen's quarter' at the center of the city, described in detail by William of Rubruck in his account of his

mission to the court of Möngke Khan in the 1250s and noted by the Soviet excavators of the site (Kiselev and Merpert 1965). Excavations not only confirmed the activity of goldsmiths and other craftsmen, but also revealed a system of furnace-fed subfloor heating systems (Bemmann et al. 2010, pp. 118-123). Combined with ceramic assemblages that corroborate Karakorum's trade linkages with China as well as the steppe, the ongoing work of the M-GKE demonstrates the potential of situating global concepts (empire, urbanism) within cultural and environmental contexts (Bemmann and Reichert 2021; Sklebitz 2018).

A phenomenon apparent in this discussion is the re-opening of sites across Eurasia that were first investigated by Russian and Soviet archaeologists, and their re-contextualization by international collaborative teams. In the case of excavations in the Central Asian former Soviet Republics, the Silk Road in many cases provides a new paradigm within which to engage with ancient and medieval urbanism, which builds upon and enriches a prior model focused on the progress of urbanization, craft production and capitalism (e.g., papers in Baumer and Novák 2019; Rante 2015). Both Marxist historical materialism and Silk Road globalization create spaces within which local processes can be compared across sites, and processes of craft specialization, urbanism, and infrastructural development can be widely discussed as emergent not only within local hegemonies, but also as articulated within global interactions. In particular, interdisciplinary work has drawn new insights from the chain of oases through Central Asia, with work focused both on irrigation and infrastructural systems, and on urban sites in Khwarezmia (Baker Brite and Marston 2013; Baker Brite 2016; Kidd et al. 2004; Kidd 2018; Negus Cleary 2013), the Bukhara Oasis (Mir-Makhamad et al. 2021), the Murghab Delta (Herrmann et al. 1993; Herrmann 1997; Simpson 2014; Williams et al. 2018) and at Otrar in

Kazakhstan (Campbell 2020; Clarke et al. 2005; Dawkes et al. 2019). These studies are complemented by ongoing investigations of highland sites, complicating understandings of seasonal movements and mobile lifeways in the context of broader Silk Road interaction in Utrushana (Stark et al. 2010), southern Uzbekistan (Bullion et al. 2022; Frachetti and Maksudov 2014; Spengler et al. 2018), and Talgar (Spengler et al. 2013). Projects working on the medieval Silk Road grapple with shared problems of both framing and reinterpretation, working between local/national historiographies and world-scale interactions. In my own collaborative project in the Republic of Armenia, we have worked to develop a dual narrative of local world-making by medieval Armenians alongside the articulation of a shared Silk Road world. A paired challenge I share with colleagues working elsewhere on urbanism, trade and complexity in the “long Mongol thirteenth century” is pushing back against entrenched local historiographic traditions based on medieval authors, which staunchly assert the erasure of local social life by Mongol hegemonic globality (Campbell 2020; Babajanyan and Franklin 2018; Franklin and Babajanyan 2018; Franklin and Babajanyan 2021; Wordsworth and Wencel 2018). The inherent interdisciplinarity of much Silk Road archaeology opens discursive space in which to unravel not only medieval and older narratives of east-west contact and conflict, but also modern models bounded by nation-states, and to re-tie connections across datasets at diverse scales.

A project paired with the investigations of mobilities and transformations of culture is the reconstruction of archaeological landscapes that supported human movement at multiple scales. This infrastructural approach has oriented archaeology spanning state boundaries in Eurasia, often in parallel with the study of transhumant and mobile polities. For instance, archaeological survey and excavation have expanded our knowledge of the networked

landscapes of the Central Asian oases that punctuated routes through Khurasan, the Iranian Plateau, Bactria, and Ferghana. Combinations of remotely-sensed work in Geographic Information Systems (GIS) with field survey and historical research have recreated networks of wells, campsites, and forts that supported transhumance and travel (Betts et al. 2009; Williams and Wordsworth 2010; Wordsworth 2015; Wordsworth 2018). A point of intersection in historical narratives, infrastructural archaeology and popular imagination of the Silk Road is represented by ongoing research into road inns, variously called *hans*, *pandoks*, *khans*, or *caravanserais* depending on their location (Figure 3). Research on caravanserais ranges in focus from architectural design and materials to the function of caravanserai systems within political structures, to the space of the caravanserai as constructive and symbolic of the wider world of the Silk Road (Campbell 2011; Franklin 2014, 2021; Franklin and Boak 2019; Tavernari 2017). Archaeological research at the infrastructural scale requires collaborative and interdisciplinary approaches, not only to collect datasets spanning national borders, but also to reconfigure archaeologies of trade and travel at scales beyond the site. The shared cultures of mobility and hospitality engendered by infrastructural spaces were themselves productive of the globality and mutuality that was perceptible to medieval and early modern travelers, and that might provide a ‘Silk Road world’ concept native to people living prior to the nineteenth-century invention of the term itself.

Silk Road Assemblages at the Molecular Scale

The challenges of scalar simultaneity posed to archaeologists by infrastructural networks are paralleled in the possibilities presented by archaeometric methodologies aimed at reconstructing movement through datasets at the scale of ions. The ongoing boom in

archaeometric innovations continues to enable new research questions related to the movement of things, organisms, and people within and across Eurasia. For instance, compositional analysis of artifacts generates quantitative data on the production and circulation of objects, which enrich previous reconstructions of trade connections and cultural influences based on typological comparisons and historical accounts. In the following section I look at a few illustrative examples of this broad body of collaborative work, which has provided insights into the circulation of technologies in textiles, glass, ceramics, stone, and metal, as well as the mobilities of humans and other organisms.

To provide an illustration of how isotopic artifact analyses have re-oriented discussions of Silk Road contact and trade, we might consider a large, accumulated assemblage of objects the findspots of which span the ‘Silk Road world.’ These objects are round mirrors cast from bronze, all of which are 15-20cm in diameter, polished on one face and often decorated on the other with reliefs of mythical creatures, hunting scenes, birds, flowers, fruit, and/or inscriptions (Figure 4). Though featuring ‘western’ motifs like grapes or languages such as Arabic, the mirrors are associated with Chinese aesthetics and production across a broad swath of time, from the 4000 BP Qijia culture through multiple phases of Han production, up until the early twentieth-century Qing dynasty (Brosseder 2015; Linduff 1996; Zhangsun et al. 2017, p. 2). The geographic distribution of these artifacts is broad, with bronze mirrors found From China to Iran, and in Bronze Age burials associated with the Bactria-Margiana Archaeological Complex (BMAC), at Pazyryk and through Semirechiye, and along the Gansu Corridor of western China (Frachetti and Bullion 2018; Kuzmina 2007). They are thus illustrative ‘Silk Road heroes,’ potentially embodying the contact and syncretism of eastern and western cultures in a single

traveling object. However, the scope of the spatial and temporal scales bound up in these small objects challenges the straightforward Silk Road story of contact, influence, and exchange between an 'east' and a 'west.' Increasing archaeometric research on the metallic composition and especially lead isotopic signatures of bronze mirrors has begun to provide resolution as well as added complexity along both chronological and spatial axes of difference. In a study focused on the later Han dynasty (AD 220-280) and drawing on thirty-four bronze mirrors from the "Silk Road terminus" of Chang'an (capital of the Western Han Dynasty, now Xi'an) located in the Shaanxi History Museum, Zhangsun et al. (2017) combined isotopic and metallurgical analysis to demonstrate the wide variability in production techniques and composition within even this relatively tight assemblage (Zhangsun et al. 2017, p. 26). But bronze mirrors can also be used as simplistic indicators of cultural-political 'impact' of Silk Road hegemonic culture on local traditions. Shifting the focus to Han bronze mirrors found in Xiongnu burials in Xinjiang, Chen et al. (2020) centered in on the proliferation of a burial ritual involving superheating and sudden cooling of mirrors that were then ritually shattered before pieces were placed in graves and potentially distributed among mourners (2020: 363). Using metallographic analysis, Chen et al. found that the practitioners of funerary rites at the Hongshangou cemetery may have practiced a local variation of the Xiongnu breakage ceremony. Based on the isotopic similarity with earlier data (including Zhangsun et al. 2017) the authors argue for a Western Han origin for the mirrors, underscoring their status as imports and arguing for a Han victory over and expulsion of the Xiongnu on the basis of this evidence.

However, ongoing work has shifted the axis of research from an emphasis on Han-era impacts to a more complex model of metalwork production and circulation. Massa et al. (2019)

completed a landmark study on early medieval (450-650 AD) quotidian metal artifacts from the site of Samdzong, in southern Tibet, where the burial assemblage from two shaft tombs included a bronze mirror and also cooking vessels, weapons, medallions, bangles, beads, and an array of horse tack (Massa et al. 2019:69-70). On the basis of portable X-ray fluorescence (pXRF) analysis and bulk component electron probe micro-analysis (EPMA), the team argued for links with South Asia as well as China and Mongolia, *but also* for more complex networks of human, metal, and object mobility to explain the intricate combinations of materials, skills, and techniques indicated by the assemblage. In particular, the constellation of techniques, metal compositions, iconographic influences, cosmological functions, ritual practices, and trade connections that resulted in the mirror and assemblages contained in burials at Samdzong (Massa et al. 2019, pp. 75-77) aptly demonstrates the myriad of scales, directions, and rates of movement necessary to frame the human-material interactions covered by the aegis of ‘the Silk Road.’ A fascinating phenomenon thus emerging from expanding archaeometric research into Silk Road exchange is mounting evidence for the decoupled circulations of object styles, materials, technologies, and cultural practices that are made even more visible by micro-scale analysis. There is a fantastic irony that many analytic techniques aimed at sourcing (i.e. localizing) objects serve rather to unravel and multiply the directionalities of movement. An object like the bronze mirror, seemingly embodying a story of ‘east-west’ exchange, instead invokes complex networks of interaction, many of which are visible only at the scale of the chemical. Beyond further troubling the Silk Road story of east-meeting west, these micro-scales also challenge the dominant story of Silk Road heroes—traveling artisans, merchants, or ‘itinerant smiths’ who single-handedly introduced the complete objects into new contexts.

Shifting scales from the object to the organism, isotopic data have transformed perspectives on the entangled life courses of humans, plants and animals within the context of Silk Road mobilities. Isotopic analysis requires complex modeling of space and time, as analysts detangle mobilities across human and animal life-courses, and over geographic distance. This deep complexity of the factors that might ultimately inform isotopic signature patterns can require acute streamlining to generate testable hypotheses, and one solution is the use of ‘the Silk Road’ to black-box some of that complexity. As an example, a recent study framed isotopic analyses of human remains in Central Asia within a provocative model, positing the medieval Silk Road as a division of labor between sedentary urban commodity producers and mobile pastoralist “operators of highland pathways” (Hermes et al. 2018, p. 1). This premise enabled the researchers to reduce potential sources of variability in the ‘isotopic signatures’ within groups to a single binary: attributed lifeways and thus role within the Silk Road model. Based on the study of carbon and nitrogen isotopic composition of human bone collagen, the authors argued for increased breadth of diet of nomadic communities between the early Iron Age and later medieval period (adding chronology back in as a variable), in support of the argument that “nomadic communities tapped into growing trade economies as agents of food exchange and broke out of insular urban subsistence food channels” (Hermes et al. 2018, p. 5). These arguments perpetuate binary oppositions between local (urban, sedentary) and global (nomadic, mobile), and thus underscore the necessity of interdisciplinarity in archaeological approaches to the Silk Road, in order to break open this black box (e.g. Spengler et al. 2018). Long-term research on the Sogdian trade diaspora, for example, is but one counter-current demonstrating the complex inter-weaving of urban and mobile practices in Central Asia (Skaff

2003; also see Fenner et al. 2020 for detailed isotopic work on ‘sedentary pastoralism’ in the medieval Mongol context). It is also apparent that molecular approaches to large-scale interactions still face challenges in pushing against, rather than confirming, Silk Road stories framed in historical sources. For instance, in looking at the relationship between Han China and the Xiongnu through the lens of carbon and nitrogen isotopic ratios, Li et al. (2020) struggled to piece together an explanatory model of changing lifeways in the Hexi corridor that counters the historiographical binarism of the Han farmers vs. Xiongnu pastoralists opposition. Promising progress is being made for earlier periods with larger datasets; Ventresca Miller et al. (2021) used extensive isotopic analysis on Scythian-era skeletal remains from Ukraine to push back against the narrative of hypermobile Scythian bands that has dominated visions of the Iron Age steppe since Herodotus.

It is interesting that so far much of the isotopic work on human mobility and diet along the Silk Road is structured by long-standing oppositions between sedentary farmers and mobile pastoralists. If archaeologists concede that the biographies of human bodies might be as complex as those of bronze mirrors, then we will need new structuring research questions for ongoing use of molecular data. It is becoming increasingly evident that the molecular datasets amassed from human and animal remains and artifacts must be contextualized within the vast molecular geography of the ‘Silk Road landscapes’ of Eurasia. For instance, in one of the studies discussed above, Zhangsun et al. (2017) point out that lead isotopes alone are insufficient to categorize metal objects by date and absolute source (*contra* Chen et al. 2020, pp. 8-9), since the environmental variation in lead isotopes along the drainage of the Yangtze River alone exceeds the isotopic variation between objects from Western and Eastern Han contexts in their

Shaanxi Museum assemblage (Zhangsun et al. 2017, p. 23). As microscopic and molecular approaches to Silk Road archaeological questions proliferate, it will remain crucial to keep the question of scale at the forefront, and to continuously interrogate the models needed to bridge the scales of isotopic signatures, human practices, and trade routes.

Bodies on the Move: Human Mobility on the Silk Road

As discussed above, a key aspect of the persisting 'Silk Road stories' is the presumption of certain kinds of protagonists or main characters. When these protagonists are embodied in people rather than cultures, then the stories tend to focus on (hyper)mobile individuals or lifeways, whose mobilities enable the dissemination of commodities, ideas and technologies. Ongoing research has enhanced the focus of Silk Road archaeology on the agencies and identities of these people; at the same time, critical frameworks continue to grapple with longstanding ideas about the relationship between mobility, agency, and gendered embodiment. Within this section I explore the intersections of Silk Road archaeology and archaeology of gender. I then further discuss the archaeology of Silk Road embodiment, gender, and mobility through two closely overlapping lenses: archaeology of transhumance and nomadism, and the applications of genetic evidence to Silk Road questions by archaeologists and interdisciplinary collaborators.

Gender and Silk Road Archaeology

The narratives of the Silk Road that inform archaeology have always been gendered. In ancient and medieval accounts, the Orient was populated with societies and ecologies that served as allegorical commentary on the norms of western Christian Europe. Fundamental in

these morally laden differences were the dispositions of bodies, relations between men and women, and overall, the politics of bodily desire. Among the gender allegories provided by Eurasia were the Amazons, who since Herodotus were employed by western writers to imagine the unimaginable: a society without patriarchy. Following in the footsteps of nineteenth-century scientific explorers who set out to confirm the reports of Herodotus, Marco Polo, and others, archaeology of the ancient and medieval Eurasian steppe has contended with constructions, deconstructions and reconstructions of gender, in particular as applied to the excavated remains of 'nomad warriors,' the embodied identity that frequently stands in as the encountering subject, the hero of Silk Road stories of movement, encounter, and transformation (Linduff et al. 2008; Linduff and Robinson 2013). In reflecting on her own methodologies, Susan Whitfield acknowledged the difficulties of making women visible through the narratives that are most frequently used to tell stories from Silk Road things (2018; pp. 9-10). Likewise, she remarked on the need to emphasize the importance of enslaved people as commodities traded along the Silk Road, underscoring that the reduction of people to the status of 'transported things' does not negate their subjectivity or agency as human beings. In her analysis of the burials of women with a combination of local and 'eastern' goods at Tillya Tepe, Robinson (2008) shifted focus from objects to persons, raising the question of the origin and possible mobility *of the buried women themselves* rather than focusing on them as end-consumers of transported goods (2008, pp. 59-60). This focus reverses the narrative of mobile, masculine merchant-travelers encountering sedentary Silk Road women, a narrative that never worked to accommodate known mobile women like Kokojin, Tamta, Khatun Bayalun, Maria

Palaiologina, or the numerous other traveling women mentioned by medieval sources (e.g. Eastmond 2017).

In a parallel development, interdisciplinary approaches to the materiality of the Silk Road continue to emphasize the importance of women as both consumers and producers of Silk Road things. To speak to the heart of the Silk Road imaginary, historians and archaeologists have repeatedly raised the importance of medieval women as the tenders of silkworms and weavers of silk (Barber 1995; Burns 2009; Linduff 2003), which though one of many commodities traded in the ancient and medieval worlds, was nonetheless a key lubricant of the machinery of beauty, power and desire in myriad, intersecting political-economic regimes (e.g. Allsen 1997; Gordon 2001). In her detailed analysis of the textiles found dressing and accompanying the Urumqi mummies contained in the Secrets of the Silk Road exhibition, Elizabeth Wayland Barber evoked teams of women working together to produce striped tabby fabrics; in addition, Barber pointed out the links between textile patterns and other mobile objects such as jewelry, and observed that fragments of cloth used in the mummies' clothing were originally from wall hangings, a central component of the portable textile architecture of the steppe (2014, pp. 34, 37; Wright 2016).

The links between the textile commodities traded along 'Silk Roads' and wider practices of emplaced world-making has been the increased focus of feminist archaeologists of technology. Francesca Bray's 1997 *Technology and Gender* demonstrated that the 'women's work' of spinning and weaving silken textiles was not only central to the division of everyday labor in the Chinese home, but was also fundamental to the sustaining of imperial cosmology and world-scale economy in the period from 1000-1800 AD. Katheryn Linduff made a parallel

observation for Shang China, linking the practice of burying carved jade silkworms and spindle whorls in the tombs of elite women to the central woman's role of weaving the fabric of a cosmologically orderly society (Linduff 2003, p. 71). In the medieval Armenian case, elite women were credited by historians with the weaving of tapestries and embroidery of lace, which were valued alongside precious metals, jewels, and imported silks for the ornament of churches (Blessing 2019; Gandzakets'i 1986). Nonetheless, the gendered aspect of commodity production as well as consumption fueling Silk Road trade tends to slip from archaeological narratives focused less on embodied divisions of labor and technologies of gendering *within* cultures, and more on an alternative axis of embodied difference and differential mobility *across* cultures, in particular between sedentary and nomadic lifeways.

Archaeology of Silk Road Mobility and Transhumance

An important framework for archaeology focused on the Silk Road is the long-standing and ongoing interdisciplinary discussion of mobile lifeways in Eurasia. Curiosity about transhumant and mobile peoples is originary to research on Eurasia, incorporating a consistent querying of the role of nomads in co-constructing the sedentary societies that historically cast them as outsiders and opponents to civilization (Khazanov 1987; Barfield 1989). In Central Asia in particular, it is increasingly difficult to consider the development of systems of irrigated agriculture, which would support oasis entrepôts and delta cities, in isolation from the transhumant placemaking practices of mobile pastoral peoples (Frachetti et al. 2017; Spengler et al. 2013; Stride et al. 2009, p. 80; but also Baker Brite et al. 2017). As the Silk Road concept has gained prominence in the last decade and following in particular the argument for "steppe roads" made by David Christian (2000), archaeologists focused on pastoral lifeways have

stepped into the fray to assert the participation, if not the centrality, of transhumant pastoralists and nomadic peoples in trans-continental travel and trade along the historical Silk Roads (Amitai and Biran 2015; Di Cosmo 2020; Honeychurch 2014a). The role of pastoralists ranges from the long-term construction of route landscapes to the navigation of mountain passes and deserts, to working as mediators, resulting in the production of hybrid material cultures aimed particularly at nomadic tastes (Wu, X., 2004). A central thread of this discussion emerged from conversations at the 2013 Society for American Archaeology meetings on the role of non-sedentary peoples in premodern globalization, compiled in an edited volume by Boivin and Frachetti (2018). For instance, Seland (2018, pp. 186-7) argues based on distributions of epigraphy and hydrology through the Syrian desert that the overlapping spheres of the premodern world-systems (*sensu* Abu-Lughod 1989) would be unthinkable without the lifeways of camel pastoralists (see also Seland 2011). Examining the early first millennium AD evidence from further east in the steppe, Brosse and Miller contend that an attention to diverse and expansive pastoralist worlds undermines a simplistic Silk Road paradigm of ‘contact,’ which portrays “global scale flows of ancient Eurasia like figurative cultural hands stretching between civilizational anchors over the barbarian realms and meeting in the merchant havens of mediating oasis states” (Brosse and Miller 2018: 163). Research centered in Mongolia and northwestern China has demonstrated the role of nomadic polities from the Xiongnu to the medieval Mongols in constructing– in unsteady collaborations with sedentary groups– the socio-political cosmologies that made space for trade and mobility across the Eurasian continent (Hoisæter 2017; Honeychurch 2014b; Miller 2014).

A co-occurring phenomenon has been that scholars working on earlier periods have extended the agency of pastoral nomads in creating the historical Silk Roads to nomadic groups living millennia earlier in the regions that would ‘become’ the Silk Road (Doumani Dupuy et al. 2018; Frachetti et al. 2017; Kuzmina 2007; Spengler et al. 2013; Spengler et al. 2018; Spengler 2020; Taylor et al. 2018). There is a rationale to some of this work on “precursor” or “incipient” Silk Roads that follows a common path of inference: 1) pastoralists and other non-sedentary peoples were crucial to the coherence of the historical Silk Roads; 2) there is evidence for pastoralism in prehistoric periods in Central Asia, ergo: 3) the pre-conditions for the historical Silk Roads (if not an actual ‘precursor Silk Road’) were created by pastoralists in prehistoric periods. While on the surface casting nomads as ‘Silk Road heroes,’ this narrative presumes (or at least heavily implies) that: a) prehistoric pastoralism was ‘prior to’ agrarian urbanism, rather than economically and politically entangled with it, and b) Central Asian pastoralism in the first or second millennium AD was a descendent continuation of pastoralism in the third millennium BC or earlier. Both of these premises have been heavily contested within anthropological archaeology (e.g. Arbuckle and Hammer 2017; Frachetti 2011; Lelievre and Marshall 2015; Spengler et al. 2021) and challenged by historians as well (Barfield 1989; Potts 2014). The ongoing conversation over modes of nomadism and migration in Eurasia and elsewhere might provide the strongest problematization of the Silk Road as a transhistorical modeling metaphor. If mobility is a complex continuum, made more complex by the differential perceptions and imaginations of space by different mobile cultures (whether seasonal transhumant pastoralists or Venetian merchants), then does an analytical distinction between mobility on- and off the Silk Road hold up, or does it collapse? As Brosseder (2015) remarks, a closer attention to the

fine grain of interaction and analysis in context “show[s] the steppe peoples’ participation in these processes and go[es] far beyond what is usually subsumed and sold under the eye- and mind-catching concept of the Silk Roads” (2015, p. 282). Critically, debates among archaeologists in Eurasia over the lifeways and ascribed identities of people engaged in non-uniform and rhizomatic interchanges of technologies and cultures form part of the context for emergent narratives of mobility and culture change framed at the genetic level.

Silk Road Genetics

The conscientious discussion of archaeological and anthropological models is of crucial importance in the rapidly developing discourse around ancient DNA (aDNA), genetics, and human and animal movements that constituted or were impacted by the Silk Road. Scalar disparity is already a long-standing challenge within genetic research; to understand genetic change a researcher must juxtapose the ‘units’ of individual organisms and species, and the chronological scales of generation, adaptation and mutation. Even more challenging are the spatiotemporal scales of relatedness, let alone mutual identification, among individuals sharing ‘lineages’ based on organellar and non-recombinant DNA (mitochondrial or mtDNA, and Y-chromosome DNA or NRY). In the last few years efforts have intensified to generate more accurate genome-scale studies (the bulk of what is discussed in this section) – though as will be apparent, the result has been a winnowing of the field down to a small number of laboratories with sufficient capacity and funding (Lewis-Kraus 2019). In the case of models of social change that draw on genetic data, or that are drawn on to explain patterns in genetic evidence, real attention must be paid to cultural explanations for population-level genomic difference over disparate scales of time and space.

Some necessary context for the intensity of interest in the *longue durée* genetic history of Eurasia is the ongoing fascination with expansive Bronze Age archaeological cultures (Yamnaya and Afanasievo in the Early Bronze Age, and Andronovo and others in the Late Bronze Age), associated by many with ‘wandering’ populations of pastoral nomads (Parzinger 2008, p. 8) and by some with reconstructed Indo-European languages such as Tocharian (Anthony 2007, see Hakenbeck 2020). Following these narratives, the arc of the Silk Road as a phenomenon of human interactions situated in the ecology and geography of Eurasia is extended, reaching a scale of millennia that is more compatible (hypothetically) with the extended timeframes of genetic change (e.g. Jeong et al. 2020). At a chronological scale this large, however, a historical phenomenon such as the Silk Road is too easily reduced to a single black box of east-west transfer, such that ‘Silk Road contact’ is presented as a straightforward explanation for the genetic intermediacy of Central Asian sampled populations ‘between’ Europeans and Asians taken as monolithic groups (Comas et al. 1998, p. 1827). At its most basic, this tendency in genetic approaches to the Silk Road to reduce Eurasian cultural and population diversity to a stark binary opposition between east and west clearly recapitulates the orientalist story of the Silk Road as a metaphor for mutual desire (e.g. Lalueza-Fox et al. 2004). In some cases this is overt, as in the work of the “Marco Polo” genetics project, which collected a linear dataset of saliva samples from “six countries dwelling along the Silk Road” and plotted them in *linear comparison* with “Europe” and “East Asia” to make an argument for “unidirectional gene flow” along the Silk Road (Mezzavilla et al. 2014). More perniciously, the power of the Silk Road as a black-boxing metaphor of change across scales enables slippages between genetic change and cultural change. While geneticists may invoke the “trading of

genes along the Silk Road" (Comas et al. 1998), the historian Peter Brown (2014) casually used genetic imagery to talk about techno-aesthetic change in the context of Greco-Bactria: "the *mutation* of the culture of the Greek colonies of Afghanistan as this culture encountered the Buddhism of northern India and reached across the high Karakorum towards China, was the *most thrilling and unexpected mutation of all of the familiar classical gene*" (2014, pp. 15-16, my emphasis). Moving in the other direction, and replacing Alexander with yet another conqueror, geneticists enacted a wild juxtaposition of timescales and agencies worthy of a medieval fabulist like John de Mandeville, arguing that a shared Y-chromosomal haplotype found across Eurasia was due to the fabled sexual conquests of Genghis Khan (Zerjal et al. 2003; but see Wei et al. 2018). The imaginary of the Silk Road designates a space-time of circulation of travelers, but also human commodities and commodified parts of humans and images of human beauty (Alram 2014; Canepa 2009; Hoffman 2001). This is especially visible in studies of mtDNA compared with those based on Y-chromosome haplogroup relatedness: while male mobility evokes images of traders, female mobility is sometimes left undiscussed (Tian et al. 2015), rather than raise the less-romantic images of bride sales, unfree labor, and displacement. A narrative of Silk Road interaction is less attractive to modern audiences if the desired commodities driving ancient economies involve human beings, but this commodification can be enclosed in the black box of the Silk Road as a genome-scale phenomenon of mobility and interaction with cosmopolitan effects (Comas et al. 1998).

The Tarim Mummies: Uncoupling Genes and Cultures

For the rest of this brief discussion of genetic archaeology I focus on the ethical implications of telling Silk Road stories with genes, and on the politics of aDNA in the ancient and modern worlds of the Silk Road. My focus here is not on the technical processes involved in aDNA genetic extraction or mapping *in the laboratory space*. But an interesting trend in molecular archaeology is the subtle extension of the black box (non-discussed aspects of research processes) to contain the construction of archaeological datasets as well as the design of what are ultimately anthropological questions to be addressed with genetic data. This is relevant because it is in these phases of research that we frequently encounter uninterrogated narratives about contact and movement in Eurasia— in other words, the same old Silk Road stories— used as metaphor-methods for pulling patterns from molecular data. This is critically relevant to the wider archaeology of the Silk Road, as more interdisciplinary archaeologists draw on aDNA for their work, and also as anthropologists raise ethical concerns about both the methods of collecting ancient and modern DNA samples and the stories these datasets are used to tell (Alpaslan-Roodenberg et al. 2021; Wagner et al. 2020).

These concerns are pertinent to the intensifying interest in using genetic evidence to reconstruct the movements of genotypic communities in ancient and medieval Eurasia. Genetic ‘communities’ are not the same as social or political communities but frequently slip and stand in for those as archaeologists and their publics look to genetic evidence to answer difficult questions about agency in the past (see Hakenbeck 2020 for a concise discussion). The chronological scope of human mobility and exchange in the case of the Silk Road adds further challenges in detangling genetics from ethnic or political identity, made even more fraught by the reinvestment of Silk Road historical groups with political meaning for reinvigorated

applications of Silk Road narratives for modern projects like the Belt and Road Initiative. To focus on a specific example, this section briefly reviews the results of genetic research into the historical Uyghurs in what is now the Xinjiang Uyghur Autonomous Region, especially connected to the Tarim Mummies. As discussed above, conflict between local Muslim, Turkic and Uyghur groups on the one hand and a series of Chinese states on the other over control of this region is older than the modern concept of the Silk Road, and the expeditions of Stein and Hedin in Xinjiang straddled the Kumul rebellion of the 1930s (Hopkirk 1980). Management and control of Xinjiang is of essential importance to the Belt and Road Initiative, as the region not only borders eight of China's Central Asian neighbors, but also possesses rich natural and cultural resources (O'Brien and Primiano 2020). An unavoidable context for this section, and for the more general discussion of archaeological genetics and the future of archaeology in Central Asia and China, is the fact that the last decade has seen a strict intensifying of state surveillance of Uyghurs in Xinjiang and the internment of more than a million Muslim Uyghurs in camps in a state effort to prevent 'radicalization' through 'reeducation' (Ramzy and Buckley 2019). It also must be stated that Chinese scientists have already been censured, and articles retracted, due to use of biological samples taken from Uyghurs without or with compromised consent. Ongoing research into the history and identity of archaeological cultures in Xinjiang must be understood as well in the context of multiple decades of Chinese censorship of Uyghur historiography arguing for the ancient presence of Uyghurs in Xinjiang, and the long-term aim of Chinese historiography to extend the antiquity of Han Chinese influence in the west as an intellectual weapon against Islamic and Pan-Turkic counter-narratives to Han hegemony (Tursun 2008: 91).

As I mentioned briefly above, the Tarim mummies are a nexus of debate over the antiquity of Uyghur settlement in Xinjiang, and the mechanisms of ancient and early medieval cultural exchange, including the spread of languages through Central Asia. The Tarim 'mummies' are a collection of desiccated corpses from multiple sites in the Tarim Basin, some of which were excavated and reburied by Aurel Stein, and which have variable dates between the second millennium BC and the fourth century AD. Good summaries of the debates over the origins and influences of the Tarim mummies are provided by Chen and Hiebert (1995) and by Thornton and Shurr (2004). These summaries present the cultures of Xinjiang as characterized by long term local continuity that is impacted ("disrupted") by Han expansion and the articulation of Silk Road Trade (Thornton and Shurr 2004, pp. 84, 88). Analyses of the DNA of modern Uyghur populations tend to map them in linear genetic space, with "Europeans" on one end and "East Asians" on the other, depending on degrees of genetic admixture (Comas et al. 1998; Feng et al. 2017, p. 2573; Xu et al. 2008). This mapping of modern Uyghurs is the framework for the analysis of aDNA from the Tarim mummies; for instance, Gao et al.(2008) situated new samples from the first-century BC Yuansha site within ancient and modern data to argue that "[c]ompared to those of the modern ethnic populations in Xinjiang, the gene pools of the ancient Yuansha, Chawuhu, Turfan, and Luobenaoer (Lop-Nor) populations have been impacted only weakly by North and East Asian matrilineal"(Gao et al. 2008, p. 212). Multiple studies of ancient and modern mtDNA however have argued for the genetic admixture of the ancient Uyghurs with steppe populations, reinforcing a model of 'Silk Road gene flow' that bundles genetic relatedness with steppic culture and Tocharian language found on documents from the Tarim (Ning et al. 2021; Xu et al. 2008). In other words, the genetic admixture of the

mummies resonates with the cultural cosmopolitanism of the Tarim material assemblages in a way that allows the residents of the Tarim to embody the essence of Silk Road mobilities (e.g. Mair 2016: 29).

However, this narrative has shifted, as Chinese scholars in particular argue for a deeper history of Han Chinese influence (at the genetic level) in the region. A team from Jilin University drew on mtDNA and Y haplogroup samples from the third millennium BC Xiaohe cemetery to push back the timeline of genetic admixture in the Tarim, arguing in particular against the standing narrative that East Asian genetic ‘impact’ in the region started only with the Han: they argue for admixture, and thus East Asian presence, starting in the Bronze Age (Li et al. 2010, p. 9, clarified and emphasized in Li et al. 2015). Fascinatingly, Li et al. (2010) argue that while the Xiaohe mtDNA samples display primarily an “Eastern Eurasian lineage,” the predominant Y chromosome haplogroup is associated with “West Eurasia” (Li et al. 2010). The Silk Road story told from mitochondrial (maternal) and Y chromosome (paternal) haplogroups is therefore gendered, a Bronze Age ‘marriage’ between West and East that clearly resonates with the dominant Silk Road narrative of both Han Chinese historiography and BRI policies.

These and other studies provide the background to the most recent and widely publicized contribution from an international team of researchers, which is based on comparing the genomes of Bronze Age mummies from three sites in the Tarim with genomic data from Bronze Age burials across the Tian Shan in Dzungaria (Zhang et al. 2021). The article argues ultimately for the autochthonous, local origin of the Bronze Age Tarim mummies; as they state, their results “support no hypothesis involving substantial human migration from steppe or mountain agropastoralists for the origin of the Bronze Age Tarim mummies, but rather we find

that the Tarim mummies represent a *culturally cosmopolitan but genetically isolated* autochthonous population” kept separate from the genomic interactions of neighboring steppe groups by, they hypothesize, the extreme environment of the Taklamakan (Zhang et al. 2021, p. 260). This uncoupling of cultural plurality from genetic admixture is important, as a call-back to basic principles in anthropology that have been neglected in the search for ‘pure’ cultures predating Silk Road contacts.

In opening the Silk Road black box of ‘east west contact’ to once again reveal long term complexity uncoupled from unidirectional migrations, we get a necessary reminder that casual use of old Silk Road stories to drive ‘hard’ science has ramifications outside of academic debate. Nearly two decades ago, Thornton and Shurr cautioned against the naive “heuristic” application by archaeologists of outdated ethnolinguistic and craniometric categories to the Tarim mummies in studies that compared them directly to the modern Uyghurs. They posited that “[f]or the modern-day inhabitants of this region, the relationship between genes, language, and culture in the past, as well as in the present, is of far greater consequence than perhaps we are willing to admit” (2004, p. 100). There is a developing concern among archaeologists that this caution should extend as well to emergent techniques in aDNA that are applied to small datasets and driven by old models. As Mallory and Mair observed in 2000, Uyghur activists seized on the compelling images of the Tarim burials, specifically naming the female mummy known as “the beauty of Kroran” (also known as the beauty of Loulan) as a “mother of [the] nation,” a claim that Mallory and Mair dismissed even while admitting it was “not without some genetic basis” (2000, p. 181). Anthropological archaeologists can affirm following the latest findings that of course, genes, language, and culture are complexly interrelated in the Silk Road

Taklamakan, as they are everywhere. Nonetheless, the identities of modern Uyghurs remain coupled to the ancient inhabitants of the Tarim through their sampled tissues, used to situate both ancient and modern peoples of Xinjiang in molecular time and in Silk Road stories. The question of a ‘genetic Silk Road’ is of existential importance for Uyghurs in Xinjiang— a concern that has yet to be robustly raised by archaeologists working on aDNA in Central Asia, despite their participation in more general calls for ethical oversight (Alpaslan-Roodenberg et al. 2021, p. 43). This debate recalls Anne Pyburn’s critical reflection on archaeological ethics as a call to care for the local over the universal: “methodologies and procedures should place immediate and local repercussions above abstract principles such as ‘the good of all human beings’ or ‘science’ writ large, or even ‘peace’ and ‘justice’” (2009, p. 162).

In summary, the dominant stories of the Silk Road continue to configure well-meaning attempts to reconstruct ancient encounters and interaction through novel methods. Ongoing work on gender by archaeologists demonstrates that there are many embodied and situated aspects of cultural transformation and transmissions that are hidden by the bias in historical sources and old models; meanwhile, the debates over nomadic and pastoral participation in Silk Road cosmopolitanism underscore the exquisite complexity of human societies and relationships in the past. A promising consensus emerging from ongoing global and collaborative work, and from discussions over the role of aDNA in Silk Road archaeology, agrees that none of these nuanced interdisciplinary discourses are done justice by the ‘Silk Road black boxes’ required to pull grand narratives from small samples of molecular data.

Silk Road Heritage

The ethical conflicts between global concepts and local stakes discussed above are of course crucial to the implications of the Silk Road as an orienting concept in global archaeological and cultural heritage. The 2008 International Council on Monuments and Sites (ICOMOS) consensus to consider transnational cultural routes as significant cultural heritage and the listing in 2014 of the first sections of the historic Silk Road as UNESCO world heritage 'sites' made unavoidable what had long been evident, that the archaeology of the Silk Road is entangled within complex narratives of the present. The power of the Silk Road as an orienting metaphor resolutely overrules any presumed opposition between an unmediated past recovered by archaeologists, and the engagements and perceptions of that past encountered through heritage and its discourses (Meskell 2015, p.2). As a narrative that makes local sites and places legible to global audiences, the Silk Road is furthermore central to nation-making and tourism brand strategies for states in post-Soviet Central Asia. Concurrent with the initial UNESCO committees on heritage routes, nineteen countries signed the Samarkand declaration for Silk Road tourism, calling for the "peaceful and fruitful re-birth of these legendary routes as one of the world's richest cultural tourism destinations" (UNWTO 1994; see also UNESCO 2002). Despite dominant narratives that portray trans-national heritage as a unifying force, Silk Road heritage and its management and tourism marketing look very different across Eurasia, from Uzbekistan with its tourism infrastructure inherited from the Soviet period and "architectural wonders" (Werner 2003) to steppe countries like Mongolia, where emphasis is put on Bactrian camels, the 'animal infrastructure' of the Silk Road sustained by pastoralist traditions (White 2020). The ramifications of Silk Road tourism for local people (aka 'stakeholders') are complex. On the one

hand, tourist expectations for 'Silk Road cities' are influenced by Victorian and orientalist accounts of Central Asia, leading to a fossilization of urban centers to preserve an imagined, authentic Silk Road past for aspirant heroic explorers in the present. But the effects are more than aesthetic. As discussed above, archaeology of nomadic polities in Mongolia emphasizes the participation of ancient and medieval pastoralists in complex sociopolitical systems; this archaeological narrative is not compatible with a romantic vision of the 'timeless nomad' that forms the basis of both tourism strategies as well as destructive state policies that relegate modern nomads to a marginalization outside global history or modern progress (Honeychurch 2010). The scalar challenges of Silk Road heritage further exacerbate the struggle of heritage universalism to tell more than a single Silk Road story.

Challenges of Deterritorialized and Global Heritage

The uptake of cultural routes as potential heritage 'properties' has led to problematization of the relationship between nation-states, cultural universalism, and material culture. The 1994 UNESCO definition of a cultural route emphasized the context, rather than the content, of route heritage, and reiterates the core scalar opposition of tangibility and mobility. The definition states that a heritage route is "composed of tangible elements of which the cultural significance comes from exchanges and a multi-dimensional dialogue across countries or regions, and that illustrate the interaction of movement, along the route, in space and time" (UNESCO 1994, annexe 1). The process of organizing Chinese nomination of routes to the UNESCO World Heritage list raised questions of how to preserve a route the physical form of which "has disappeared— assuming it ever existed" (Feng 2005, p. 1). In many ways the epistemological work to define the object of protection in the case of a cultural route reveals

fundamental ambivalences at the core of the heritage concept, corroborating ongoing conversations within heritage archaeology over the unwieldiness of the ‘monument’ as a concept (e.g. DeSilvey 2017). For instance, the findings of early UNESCO work to define what would count as a route in China resulted in the following bundled definition:

“[A] cultural route could be defined in terms of *space* (the route ran through sites, monuments, constructions, buildings, ways and areas of influence), *time* (the beginning and end of its use, it’s frequency, intensity and variations) and *cultural criteria* (impact of spiritual and/or material exchanges, impact on human memory or experience impact of the volume and nature of the exchanges)” (Feng 2005, p. 2, my emphasis).

This definition in effect constructs another conceptual black box, allowing the mechanisms of the Silk Road (sites and routes, practices through time) to be synonymous with its “impacts,” collapsing causes and effects. UNESCO convened the International Symposium on the Silk Roads in Xi’an in November 2002; the subsequent ICOMOS Xi’an Declaration of 2005 made a particular contribution to the discourse by overall emphasizing “setting,” defined as “the immediate and extended environment that is part of, or contributes to, its significance and distinctive character.” (ICOMOS 2005, p. 2). The declaration reveals in more detail a key challenge of global heritage, or heritage in global context: that preservation and development are frequently conceived of in opposition to one another, putting institutions aimed at archaeological conservation in the position of having to choose between past and future.

All of this speaks to a movement of approaches to heritage away from the isolated 'monument' and towards a conception of cultural heritage as networked, spatiotemporally complex and situated in practice as well as in place. This movement overlapped with an increasing sense of urgency in filling in the gaps in the world map of World Heritage listings, in particular the large blank space of Central Asia (UNESCO 2008). But how do states and state institutions preserve such deterritorialized heritage conceived at disparate scales? An immediate resource is apparent in digital approaches to Silk Road heritage, using media such as Geographic Information Systems (GIS) that are suited to the managing of point as well as linear data and to representation at multiple scales. An ICOMOS commission funded by Chinese donations to the World Heritage Fund and supported by UNESCO initiated a multi-year data gathering and research project in GIS. The resulting Thematic Study explores many of the challenges to the project; even at the scale of the 'whole Silk Road,' the project struggled to define edges for the datasets (Williams 2014, p.12). The study also raises the critical issue that continental-scale projects are still dependent on the volunteering of data by individual member countries (Williams 2014, p. 71).

The designation of Silk Road routes as universal cultural heritage is fraught with contradictions, as the mechanics of UNESCO membership and compliance is ultimately in the hands of nation-states and state institutions. The cooperation and cosmopolitanism implied in the rhetoric of serial nominations are at odds with ideas of heritage fundamental to state- and nation-making that center locality, autochthony, and/or nativity (e.g. Zhu 2017). These are critical contradictions for archeologists working on and in the worlds of the Silk Road. Anne Pyburn (2009) points out a deep incompatibility between the depictions of boundless

premodern mobility associated with the Silk Road, and the need of nation-states and nationalist movements such as that in Kyrgyzstan for monuments that measure up to urban, western-oriented expectations (see also Levin 2018). Or as Tim Williams put it: “clearly the Silk Roads do not represent the same historical/cultural group, or the same type of property in a geographical zone: indeed the Silk Roads are an anathema to both these concepts” (2014, p. 3). At the same time, potential members of Silk Road heritage initiatives must choose between local histories and global heritages; as Tim Winter observed, the pressure of participation in world heritage “encourages states to be internationally disposed, wherein a culture of cooperation propagates internationalized cultural nationalisms and the building of bridges through shared pasts” (2016, pp. 27-8). For many, the absorption of local history into global dialogue raises deep anxieties of identity. Certainly, in the Republic of Armenia a narrative of mobile and ‘promiscuous’ Silk Road culture does not do the work required of heritage by a nation that perceives itself under existential threat; the failure of Azerbaijan to nominate medieval Christian heritage in the contested enclave of Nagorno Karabakh to UNESCO World Heritage designation is perceived by Armenian commentators as an urgent crisis not just of national self-determination but of survival (Ambartsumian 2021). For interdisciplinary scholars working on Silk Road heritage in the Caucasus and elsewhere, the balancing of local histories and global narratives is central to the agenda and obligations of future research.

Figure 6. The Ming Dynasty (fourteenth century AD) Great Mosque of Xi’an, a UNESCO World Heritage Site under the Silk Roads Program (Adapted image CC BY-SA 4.0).

Silk Road Archaeological Heritage and the Belt and Road Initiative

The emergent paradigmatic illustration of these contradictions is the conjoined cultural and geopolitical emphasis on Silk Road heritage woven through the PRC's Belt and Road Initiative. I will keep my discussion of the BRI relatively brief, as it is an enormous topic that has received detailed scholarly attention (see Whitfield 2020; Winter 2019). The Belt and Road Initiative (also called at various times the One Belt One Road or New Silk Road program) is a Chinese geopolitical strategy with global impacts and transnational interests; the rhetoric of this program draws on affective invocations of the Roman-Han and (to lesser extent) medieval Silk Road world and their (modern) associated narratives of perpetual peace and *doux commerce*. Perhaps most important for archaeology of the Silk Road in China as well as in countries of Central Asia influenced by the 'geocultural power' (to use Tim Winter's term) of the BRI is the hegemony of metaphor enacted in Chinese deployment of both of our Silk Road stories (the hero's journey and the mutual gaze), as techniques of making future worlds and future politics. To put this in more familiar terms, the BRI draws upon and deploys powerful and nuanced programs of so-called soft power (e.g. Qian 2022) aimed at extending a singular national past into a global future (Shelach-Lavi 2019, p. 25).

I do not have space here to explore the entirety of China's heritage landscape, wherein Silk Road heritage intersects in complex ways with the construction of local tourist attractions and national landmarks. This section highlights ongoing work that illustrates points of intersection between nation- and material memory-making in China, as an example of tensions within Silk Road heritage more broadly. A critical theme of archaeological heritage conservation and representation in China is a different approach to the concept of 'authentic materiality;' as

has been extensively explored by Yujie Zhu, there is an ongoing debate among heritage practitioners in China over the culturally appropriate translation of the imported (and very much freighted) European term *authenticity*. As Zhu argues, an emergent appreciation of the ‘old’ or traditional in China has developed in tandem with critique of a universal European definition of an authentic or original state that is the goal of reconstruction or preservation (Zhu 2017; see also Nakano and Zhu 2020, Su et al. 2020). In a mode familiar to scholars of heritage in the ‘west,’ the preservation of Silk Road heritage in China thus represents negotiations of archaeological materiality with the needs of local people for sites of nostalgia, as well as national motives to extend narratives of political identity into archaeological pasts. These negotiations of material ‘truth’ are the context for the use of heritage to situate China’s relationships with minority populations, and especially Chinese Muslims. This is especially visible in the western city of Xi’an (historic Chang’an), designated as the Chinese terminus of the Silk Road (Pang 2014). Since 2005, the Xi’an municipal government has worked to reconstruct the central urban fabric through the Tang Imperial City Renaissance Plan, aimed at ‘returning’ Xi’an to its form as the seventh-to-tenth-century trade capital of the Tang Dynasty (Feighery 2011, p. 39). After the designation of Xi’an as a UNESCO Silk Road Heritage site in 2014, special attention was given to the preservation of the Lianhu Muslim Quarter, a mile-square site inhabited by thirty thousand Hui Muslims (Zhu 2020, p. 7); the focus on this site had clear geocultural implications in ongoing Chinese negotiations with Central Asian partners in the BRI. Heritage sites like the Xi’an Muslim Quarter continue to be sites of conflict over the narrative of China’s Silk Road history, as the municipal government’s desire for a ‘medieval’ ethnic quarter with picturesque mosques (Figure 6) and ethnic street food clashes with the

needs of a living Muslim community who sees their future at risk as traditional homes are torn down to make space for curated heritage tourism (Feighery 2008: 328; Feighery 2011; Zhu 2019). The use of modern Hui Muslims as set-dressing for a state-approved vision of medieval multiculturalism along the Silk Road encloses the Muslim communities of the ‘western regions’ in the medieval past, cordoned off from both China’s present and its future.

The negotiations of BRI futures and archaeological pasts are of course also visible at sites ‘along the road,’ in regions rich both in Silk Road archaeology and tactical and material resources needed for BRI expansionism. The violent legacies of orientalism and Great Game politics in Central Asia also challenge the continued extension of utopian Silk Road narratives invented by those same politics to the inhabitants of nation-states created in empires’ wake and subject to the reactivation of those narratives by the BRI. The dominant example of this is Afghanistan; renewed archaeological research following the 2001 US invasion expanded an already immense record of urbanism, infrastructure, architecture, and artistic hybridity that attests to the rich history of encounter, exchange and cultural production in river valleys, oases, and plains (see Ball and Hammond 2019 for a review; Mairs 2020; Murdock and Hritz 2013; Thomas 2017; see also Flood 2009). The site of Mes Aynak, an early Buddhist monastic center in the central mountains of Afghanistan has emerged as a frequently cited example of endangered Silk Road heritage (Klimburg-Salter 2018; Lawler 2011). A well-preserved monastery that sits atop a seam of copper worth \$50 billion, the site straddles multiple and contradictory citations of the Silk Road as linked to the prosperity and dignity of the post-imperial nations of Central Asia. As a Buddhist monastery it represents the cosmopolitanism and perceived hypermobility of the early medieval Silk Road— and serves as a counterpoint to the perceived intolerance and

“lack of cultural accomplishments or abilities”(Benard and Sugarman 2013, p. 75) of the Taliban regime, known for their destruction of UNESCO sites such as the monumental Buddhas at Bamiyan (Flood 2002). But the site is also threatened by the same infrastructural and development initiatives that are supposed to bring prosperity, modernity, and new connectivity (cosmopolitanism) to places like Afghanistan under the auspices of the new Silk Road. As heritage research has increasingly emphasized, development and urbanization are leading threats to archaeological heritage, outstripping deliberate terrorist destruction by a wide margin (Hammer et al. 2018). But at Mes Aynak these categories are fuzzy. The mining initiative at Mes Aynak, MCC, is centrally implicated in the Belt and Road initiative as one of the primary development arms of construction and extraction (Petkova and van der Putten 2020). In the wake of the American defeat and withdrawal in 2021, MCC and mining interest Jiangxi have expressed intent to continue working with the Taliban government to develop copper extraction from Mes Aynak, further undermining the premise that the hegemonic narratives of BRI and UNESCO heritage are deploying the image of the Silk Road in aligned ways.

In summary, the nearly two decades of discussion on how to conceptualize, designate and represent Silk Road heritage reveals parallels to the challenges and opportunities of deterritorialized Silk Road archaeology more generally. At the same time, Silk Road heritage reproduces many of the tensions between universal/global and particular/local that run through the interlinked concepts of world- and national archaeological heritage. In the context of heritage as a tool of diplomacy, development and ‘geocultural power,’ uninterrogated Silk Road stories easily become hegemonic as implements of state power and nation making, even

as the material evidence for transnational complexity continues to inspire curiosity and wonder on the part of global and local publics.

Conclusions

In this brief review I have tacked back and forth between Silk Roads invented and reinvented, between the rich contributions to archaeologies of globality and globalization enabled by Silk Road frameworks, and the methodological challenges presented by the Silk Road as a metaphor for contact, encounter, and transformative impact. I began by situating the construction of the Silk Road in its historical context, as a concept that facilitated the knitting together of material evidence for mobility, exchange, and influence by archaeologists in Central Asia, even as the resulting narratives themselves were used to tighten the seams of European and Chinese imperial projects. These late-nineteenth and early-twentieth-century beginnings in turn contextualize the diverse projects of the last decades that have worked to re-invent the Silk Road as a framework for thinking about world-scale systems, human and material mobilities, and processes and experiences of globalization at different scales. The wide appeal of the Silk Road as an imaginary has created public and academic interest in sites and regions previously consigned to area studies and has enabled new and productive interdisciplinary and transnational collaborations that work simultaneously at the scales of the local and transregional. Likewise, the framing metaphor of the Silk Road has broken down categories within archives and museums, enabling new juxtapositions of texts and objects, and new techniques for interpreting artifacts in themselves.

As I explored, archaeologists working on Silk Road questions have reconfigured anthropological approaches to human mobility, whether through transhumance, travel,

migration, or trade. The question of gender as a cultural orientation in the world has been central to Silk Road archaeology; the circulation of material culture created common cultures of embodiment and bodily power across Eurasia, and ancient and medieval ideas of gender were fundamentally shaped by the differential mobilities and agencies of people within landscapes and between emergent categories of identity and belonging. Likewise, the archaeology of the Silk Road is intertwined with archaeological research on transhumance and nomadism, which has in turn transformed our awareness of how human participation in more-than-human ecologies shapes not only landscape, but possibilities of movement and expanding borders of historical worlds. New techniques for tracing the movement of humans, animals and plants—such as isotopic and genomic analyses—add rich depth to ongoing discussions of human mobility in Eurasia; my critique of Silk Road black-boxing within genetic research in Central Asia and Xinjiang in particular is tempered by my regard for the rigor and reflexivity of archaeologists who have been working on these questions for decades.

Finally, I briefly discussed some of the possibilities of Silk Road heritage, both as an avenue for transnational collaborative storytelling and as a powerful tool for narrative nation-making. The tensions between globality and locality inherent to Silk Road heritage compound the preexisting oppositions between universality and particularity at the heart of cultural heritage as a practice and paradigm. As an accumulating scholarly discourse has emphasized, these tensions are especially (but in no way exclusively) apparent in the cultural-political aspirations of China's Belt and Road Initiative. My review underscores an observation made by others, that the 'soft power' of Silk Road heritage reinvents and redeploys the entangled imperial and intellectual projects built into the Silk Road from its conception (Chin 2013;

Whitfield 2020). The Silk Road thus provides a rich case study for interrogating modern heritage possibilities, as well as a key example of the power invested into transforming national destinies into global heritage futures.

As an analytical framework, the Silk Road has facilitated interdisciplinary, transregional, multi-sited and collaborative archaeology. Developing in tandem with critical archaeologies of globality and globalization, engagements with the Silk Road narrative can push critical questions of perception as well as imagination, which challenge the scales of archaeological models. To put that more concretely, an ongoing challenge for archaeologists who study local participation in large-scale phenomena— whether ‘horizons,’ ‘ecumenes,’ ‘world-systems,’ or ‘hegemonies’— is reflecting on how (or if) people in the past perceived the structures that we see as structuring their political destinies and daily lives. An early question I received about my research in Armenia was “did anyone living in this valley care that they were on the Silk Road?” This question of effective vs. affective participation in a world scale culture (i.e. the difference between doing and perceiving) has driven my— and other colleagues’— research towards interdisciplinary attempts to resolve these questions.

The question of perception and participation also underscores the difference between the world(s) made by trade, exchange and mobility in the ancient world, and the world we think of when we hear ‘the Silk Road.’ From the perspective of medieval archaeology of the middle east and Central Asia, the Silk Road has provided a paradigm that transcends ‘Islamic’ or ‘Byzantine’ archaeology and pushes for the study of materiality and ideas in motion, of emergent co-participation in shared global cultures as well as the active fashioning of situated traditions from global fabrics (pun intended). Likewise, discussions of the Silk Road have

integrated archaeology in Asia within world- and even European archaeological conversations. A vast body of work that was barely touched in this review is the archaeological and historical work on mobility and exchange in the Indian Ocean, including the Persian Gulf and African coast, which has already begun to break away from the framing of 'maritime Silk Road' under which it gained prominence. I predict that as collaborative and interdisciplinary work increases, new narratives of collaborative world-making will turn the narrow east-west binary of the Silk Road inside out. This can only be hastened by post-colonial, symmetrical and collaborative archaeology that multiplies the potential narratives and metaphors in our toolkits.

In their attempts to construct narratives of Silk Road culture at the global scale, archeologists share a project of world-making with the creators of many of our historical sources. A technique we share with ancient and medieval geographers, cartographers, clerics, and historians is allegory: a mode of metaphor wherein the particular stands (frequently on figural feet) in for the abstract. "The Silk Road" is such an allegory, and a robust one—understood as containing wide diversity in perception, agency and contingency in figural shorthand as a road from east to west. We tend to expect that our publics will understand the metaphor as what it is, to understand that there wasn't actually a single road, that not even Marco Polo traveled the entirety of it, that material, linguistic or genetic 'encounters' took centuries, and were perhaps never even perceived as encounters by the peoples in question. As I have explored, this richer narrative is lost in analyses that use the Silk Road as a methodological black box, reducing temporal, spatial, and cultural complexity to linear east-west transfers or unidirectional impacts of the global on the local.

But in addition to these challenges, there is real promise in the compelling efficacy of the Silk Road as an allegory for thinking globally in the ancient world. In navigating the emergent de- and reterritorialization of historical narrative, of material metaphor and technological transformations of the world required by Silk Road archaeology, we ultimately reflect on how people in the past imagined phenomena at scales larger than their own homelands, and of greater duration than their own lifetimes. We also reflect on techniques by which people provincialize the global, potentially enwrapping unthinkable spans of space and time within the undiscussed everyday. These remain critically relevant, as tools developed by archaeologists with wide applicability for negotiating uncertainty and scalar agency in the present and future.

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Figure Captions

Figure 1. Map showing the general extent of Silk Road Eurasia and sites discussed in the article.

The white dotted-line rectangle indicates the extent of Figure 5. Map created by the author.

Figure 2. Sir Marc Aurel Stein's excavations of a house at Niya, in the southern Tarim Basin of Xinjiang. From Stein, M.A. (1912). *The Ruins of Desert Cathay* Vol. 1, Pl. 84. (Public Domain).

Figure 3. Interior of an Ilkhanid Mongol period (fourteenth century AD) caravanserai in the Selim Pass, Armenia. Photo by the author.

Figure 4. Medieval bronze mirrors demonstrating the 'cross-pollination' of techniques, motifs, and styles. Left-hand mirror: Tang China, eighth century AD, Diameter 17.1cm; Metropolitan Museum of Art 25.82.8 (Adapted from Public Domain image). Right-hand mirror: Iran, twelfth

century AD, Diameter 19.4cm; Metropolitan Museum of Art 42.136 (Adapted from Public Domain image).

Figure 5. Map of the Tarim Basin showing sites discussed in the article and cited literature. Map created by the author.